

The output in Figure 5 shows what happens when an Ncat connect mode session is used to communicate with the preceding listener.

```
[root@ipa ~]# nc 192.168.123.2 4343
ls
1
anaconda-ks.cfg
ssl-build
pwd
/root
ll
█
```

Figure 5: The Ncat connection

Each line of text sent to the server is executed by `/bin/bash`. The resulting output of the commands is sent back to the network client.

Note: Ideally, `-e` is not the recommended option as the shell should never be directly connected to a network port for obvious security reasons. I have just shown it to explain the feature of the `-e` option.

Ncat supports both IPv4 and IPv6. The `-4` and `-6` options force Ncat to use either IPv4 or IPv6 respectively.

Monitoring network traffic

IPTraf is a network monitoring software originally developed in the mid-1990s. This tool can be installed by downloading the *iptraf-ng* package.

The *iptraf-ng* command launches the application. The program requires super user access, so it must be executed by the root. *IPtraf-ng* has a curses menu interface. The screen shot (Figure 6) shows the main menu that is displayed when the tool is first launched.

Selections are made with the up and down arrow keys; then, pressing *Enter* executes the current selection. Alternatively, typing the highlighted single character can also execute a command.

IPtraf-ng can monitor the current network connections. It can display detailed or summary counts about the network



Figure 6: The *IPTraf* menu

interfaces on the local system. It can also display UDP and ICMP packet information.

The *Filters* main menu selection allows an administrator to create filters to include (or exclude) specific types of network traffic. Each filter is a collection of rules that can select packets based on the source (or destination), address, port, and IP protocol type.

This article has only covered how to test network connectivity and some important networking tools. It is the first part of a series on troubleshooting network issues. **END** 🐧

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